Build the full prediction

Mission control wants force → acceleration → velocity → position.

20 A

THE SCENARIO

A 4 kg cart starts from rest at x = 2 m. A 14 N pull acts right; friction is 2 N left. The forces act for 3 s.

LINK 1 · NET FORCE

$\Sigma F =$ ____ N

LINK 2 · ACCELERATION

$a = \Sigma F \div m =$ ____

LINK 3 · VELOCITY

$v = v_0 + at =$ ____

LINK 4 · POSITION

$x = x_0 + \frac{1}{2}(v_0 + v)t =$ ____

RANGE

The model bound is ±1.5 m. Report: ____ m to ____ m. Guaranteed? ____

WHAT IF…

…friction rose to 14 N? Which link changes first and what happens?

ANALYST SIGN-OFF

The link I trust least is ____ because ____. For 2026-XJ, we do the east and north predictions separately because ____.

SYNTHESIS 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

VOL. 1 · NO. 20 · SYNTHESIS REPORTER: _____

The 2026-XJ Daily

Cart prediction confirmed — or was it?

BREAKING · HEADLINE (include the range)

THE STORY

Explain how the team got from forces to a final position. Name each link in the chain.

BY THE NUMBERS

ΣF = ____

a = ____

v = ____

x = ____

range ____

LETTERS TO THE EDITOR

"The cart has zero net force, so it can't move." — Confused in Room 108. The editor replies:

SCIENCE DESK

Zero net force, already moving at +2 m/s for 3 s: velocity and displacement?

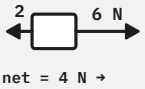
CORRECTION BOX

Yesterday we printed "5 m — guaranteed." Write the correction:

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: the chain

Language goal: I can **connect** cause and effect.



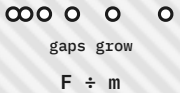
net force

fuerza neta · força resultante

all forces added up

my language: _____

use it: _____



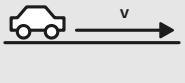
acceleration a

aceleración · aceleração

how velocity changes

my language: _____

use it: _____



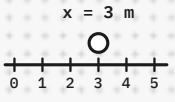
velocity v

velocidad · velocidade

how fast + which way

my language: _____

use it: _____



position x

posición · posição

where it is

my language: _____

use it: _____



cause

causa · causa

what makes something happen

my language: _____

use it: _____



effect

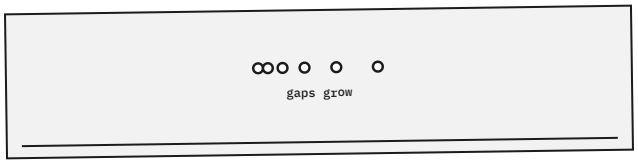
efecto · efeito

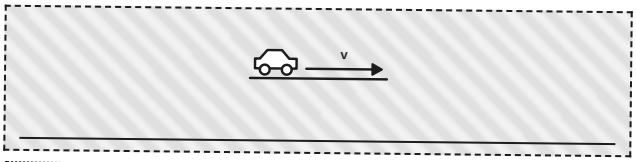
what happens because of it

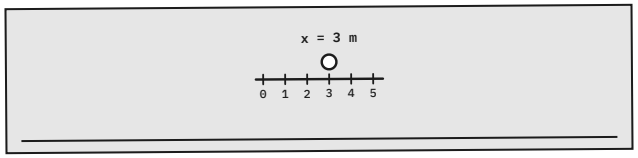
my language: _____

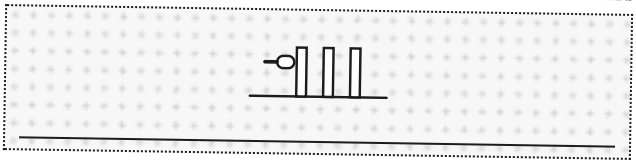
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner □

SYNTHESIS 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

ANTOCCI ENGINEERING · DWG U1-20 · SYNTHESIS

DELIVERY DRONE: LANDING SEQUENCE

A 2 kg drone descends. Up = +. Plan the chain from forces to landing.

CALC 1 · NET FORCE

Thrust 16 N up, weight 19.6 N down. ΣF = ____

CALC 2 · ACCELERATION

$a = \Sigma F \div m$ = ____ m/s²

CALC 3 · VELOCITY AFTER 2 s

Starts at 0 m/s. $v = v_0 + at$ = ____

CALC 4 · DROP IN 2 s

$\Delta y = \frac{1}{2}(v_0 + v)t$ = ____

DRAWING · FBD + v–t SKETCH OF THE DESCENT

ENGINEER'S NOTE

What thrust would make the drone descend at a constant velocity? Explain with the 1st law.

SYNTHESIS 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 20 · SYNTHESIS · SEI · SENTENCE FRAMES

Say it, then write it: the chain

Language goal: I can **sequence** cause and effect.



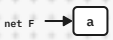
cause
causa · causa
what makes it happen



effect
efecto · efeito
what happens



then
luego · então
next in order



so
así que · então
shows a result

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: force · acceleration · velocity · position · range

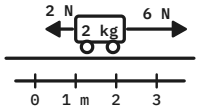
A net ____ causes an ____. Then the ____ changes. So the ____ changes too. We report a ____ because the model is not perfect.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The net force is ____, so the acceleration is ____ .
Then the velocity is ____, so the position is ____ .

3 · SOLVE WITH THE STEPS



2 kg cart, 6 N right, 2 N left, from rest at 1 m, for 2 s.

STEP 1 · ΣF AND a
____ N · ____ m/s²

STEP 2 · v
0 + ____ × ____ = ____

STEP 3 · x
1 + ____ = ____ m

4 · EXPLAIN IT · CHOOSE A FRAME

☐ First ____, then ____, so ____ .
☐ The cause is ____ and the effect is ____ .
☐ We give a range because ____ .

I said it ☐ · I wrote it ☐

SYNTHESIS 1 2 3

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Fix the group chat

20 R

DIRECTIONS · Reply in the empty bubbles. Show the math on the right.

XJ-7 tracking crew

NADIA

2 kg cart, 6 N pull, 2 N friction, from rest. a?

YOU ①

ELI

cool so after 2 s it's at 4 m/s

MAZ

how far did it go in those 2 s?

YOU ②

ELI

it started at 1 m. so it's at exactly 5 m, guaranteed

YOU ③

NADIA

and if the net force were zero?

YOU ④

SHOW YOUR RECEIPTS

① ΣF → a

② Δx = avg v × t

③ x_0 + Δx, with a range

④ ΣF = 0 →

SYNTHESIS 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else